SHENG-HAO (JERRY) WU

Senior Software Engineer - ML Compilers, Inference Accelerators & Edge Systems
Portfolio | LinkedIn | GitHub | iamjerrywu188@gmail.com | 412-680-6223 | Santa Clara, CA

SKILLS

ML Compilers : MLIR (Dialects & Passes), Graph IR Transforms, Compiler Auto-Tuning, DAG Scheduling, Graph Partitioning, Operator Lowering & Decomposition, Memory Planning, PyTorch, ONNX

HW Acceleration : NPU/VPU Architecture,, Edge & On-Device AI Inference, Vision Models & LLMs, Quantization & Reduced-Precision (MX/BF16/INT8), HW/SW Co-Design, Cycle Simulators, HIL Silicon

Languages & Systems : C/C++, Python, Linux Kernel, RTOS, Concurrency & Synchronization, DMA Memory Hierarchy, GoogleTest, LCOV, CI/CD, GDB

WORK EXPERIENCE

Ambarella Inc. Santa Clara, CA | Feb 2023 - Present

Senior Software Engineer

- **Next-Gen Transformer & LLM Acceleration**: Architected the transformer compilation pipeline for edge LLM inference in an MLIR-based compiler; guided hardware–software co-design via cycle modeling and built PyTorch reference models to validate reduced-precision parity (MX/BF16).
- **Compiler Auto-Tuning Framework (VAS Explorer):** Architected VAS Explorer, a parallel compiler auto-tuning system searching graph partitioning and DMA schedules with shared-IR reuse; achieved >5% latency speedups on 60%+ of vision models using multi-fidelity cost models (2,000+ stress runs).
- **Graph Transforms & Memory Planner Integration:** Implemented multi-axis graph transforms for oversized tensor ops and hardened compiler/memory-planner integration to maintain graph-ordering consistency; eliminated metadata overflows and unnecessary partition cuts across ~1,360 regressions.
- **Hardware-Aware Operator Lowering:** Built hardware-aware lowering for convolution, histogram, bucket-sort, and RNG primitives under accelerator limits; designed a targeted histogram decomposition pass delivering 40–60% runtime acceleration on benchmark vision models.
- **Hardware-in-the-Loop (HIL) & Silicon Codegen:** Co-developed a silicon codegen and Hardware-in-the-Loop (HIL) backend translating graph IR into on-device runtime images; automated physical execution across NPU boards, eliminating memory-stride and tensor-layout mismatches.
- **CI Automation & AI Triage Assistant:** Engineered CI regression pipelines across ~10 hardware targets cutting model regeneration 2.75x (22h to 8h); developed an automated compiler bisection engine isolating commit failures for an in-house AI triage assistant
- **Compiler Quality Gates & Testing Infrastructure:** Established compiler unit-testing quality gates using GoogleTest across hardware targets; integrated LCOV coverage and complexity tracking to guide pass refactoring, lifting core coverage to 87% line and 83% function

Meta Platforms, Inc. Menlo Park, CA | May 2022 - Aug 2022

Solutions Engineer Intern

- **Distributed Workflows & Telemetry**: Engineered asynchronous backend pipelines and decoupled platform workflows for “Meta for Business," improving system reliability, telemetry tracking, and multi-tenant performance at scale

Mediatek Inc. Hsinchu, Taiwan | Sep 2016 - Jun 2021

Senior Software Engineer (promoted from Software Engineer)

- **Leadership & Multicore DVFS**: Led engineering team developing modem DVFS and power-saving drivers on Nucleus RTOS across multi-core SoCs, reducing power margin by up to 25% under heavy throughput.
- **Concurrency & System Profiling**: Architected thread-safe APIs and critical-section synchronization to eliminate kernel race conditions across multi-core subsystems; engineered offline trace analyzers profiling scheduling latency and lock contention.
- **Silicon Bring-Up & Commercial Scale**: Executed post-silicon bring-up and low-level debugging (JTAG, logic analyzers) for baseband chipsets shipped in millions of commercial devices, establishing test suites achieving 90%+ code coverage.

EDUCATION

Carnegie Mellon University - M.S in Information Networking — Pittsburgh, PA | Sep 2021 - Dec 2022

National Cheng Kung University - M.S in Electrical Engineering — Tainan, Taiwan | Sep 2014 - Jun 2016

National Cheng Kung University - B.S in Engineering Science — Tainan, Taiwan | Sep 2010 - Jun 2014